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Tutorial 6

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1)

Let A be the total number of possible samples:

$$A = \binom{100}{3} = C(100,3) = \frac{100!}{(100-3)!\,.\,3!}$$

Let B be the number of ways to choose 3 questio that he knows the answers:

$$B = \binom{90}{3} = C(90,3) = \frac{90!}{(90-3)!\,.\,3!}$$

Let C be the event that the student will pass the examination:

$$P(C) = \frac{B}{A} = \frac{\dfrac{90!}{(90-3)!\,.\,3!}}{\dfrac{100!}{(100-3)!\,.\,3!}} = \frac{90.89.88}{100.99.98} = \frac{178}{245}$$

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2) $\binom{n-1}{k} + \binom{n-1}{k-1} = \frac{(n-1)!}{(n-k-1)!\,k!} + \frac{(n-1)!}{(n-k)!\,(k-1)!}$

$= \frac{(n-1)!\,(n-k)}{(n-k)!\,k!} + \frac{(n-1)!\,k}{(n-k)!\,k!}$

$= \frac{(n-1)!\,[(n-k)+k]}{(n-k)!\,k!} = \frac{(n-1)!\,n}{(n-k)!\,k!}$

$= \frac{n!}{(n-k)!\,k!} = \binom{n}{k}$

3) The light bulbs are defective:

$1000 \cdot 2\% = 20$ (light bulbs)

The number of ways to buy 2 faulty ones and 18 not defective bulbs from 1000 bulbs is:

$C(20,2) \cdot C(1000-20,18) = A$

The total number of samples is:

$C(1000,20) = B$

The required probability:

$P = \frac{A}{B} \approx 0,0519$

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4) The number of ways to choose full meal

$$5 . 3 . 7 . 6 . 8 = 5040$$

The number of ways to choose a meal if either an appetizer or a soup is ordered, but not both: $(5+3) . 7 . 6 . 8 = 2688$

5) The number of ways to choose 6 people sit in 6 seats in a line at cinema:

$$P(6,6) = 6! = 720$$

The number of ways to choose 6 people sit around a dinner table:

$$1 . 5! = 120$$

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6. Let A be the number of ways of choosing 5 letters from the alphabet.

Let B be the number of seconds in a year.

- $A = C(26,5) = 65780$
- $B = 31\,536\,000$

$\Rightarrow$ The statement is false $(65780 < 31\,536\,000)$

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